



Design and Implementation of a Database Management System for Crop Planning and Farm Input Management System

DBMS Project-Based Learning — Review 1

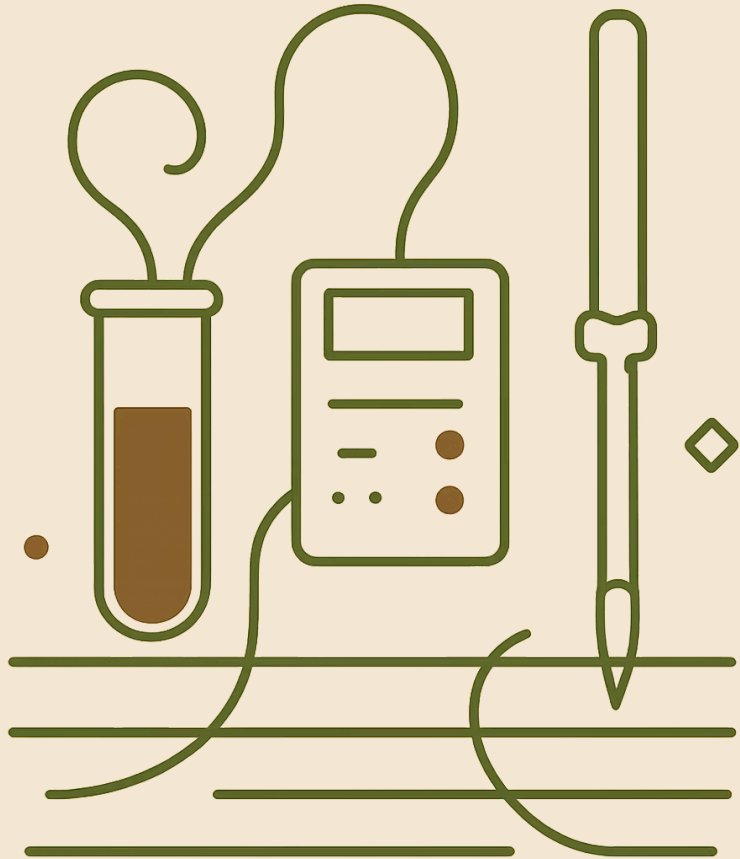
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Problem Statement

- Farm records are fragmented or manual: fields, seasons, crops, soil, irrigation, inputs, labour, harvests, and sales.
- Hard to compare cost vs. yield or make data-driven planning decisions.
- Need a unified, queryable system to support operational and planning workflows.



Objectives

Normalized DB

Track full crop-to-sale lifecycle in a relational model.

Cost vs. Yield

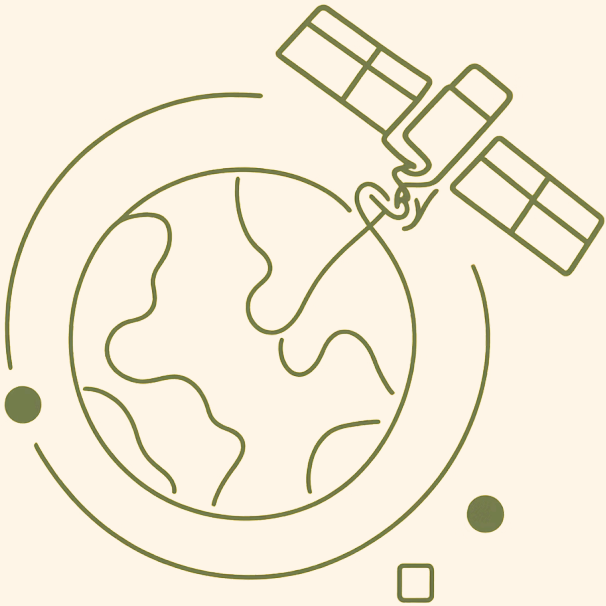
Enable profitability and input-efficiency analysis.

Decision Support

Guide inputs, irrigation, and labour planning.

Front-end

Provide day-to-day data entry & reporting (UI in later reviews).



Scope

System covers farms, fields, seasons, crops & varieties, soil tests, crop plans, inputs & applications, irrigation events, labour activities, harvests, buyers, and sales.

Implementation targets: MySQL / PostgreSQL / Oracle. Front-end: Java or Python (CRUD & UI in later reviews).

Target Users



Farm Owners / Managers

Strategic planning, profitability analysis.



Field Supervisors

Daily data entry and task coordination.



Agronomists / Soil Advisors

Soil test interpretation and recommendations.



Labour & Sales Staff

Manage activities, harvest records, and buyer relations.



Functional Requirements

- Crop planning and crop-plan lifecycle management
- Soil-test entry & tracking
- Irrigation scheduling and event logging
- Input issuance and application records
- Labour activity logging
- Harvest recording, inventory, and sales tracking
- Reporting: field plans, input use, yield, and profitability



Key Entities (13)



Farm

Primary
owner/operation unit



Field

Parcel with
boundaries and soil
attributes



Season

Crop calendar
window



Crop / Variety

Species and cultivar
details



Soil Test

Lab results and
recommendations



Crop Plan

Planned inputs,
schedules, and
targets



Input / Application

Fertilizer, pesticide,
seed, etc.



Irrigation Event

Watering
occurrences and
parameters



Labour Activity

Tasks, hours,
personnel



Harvest

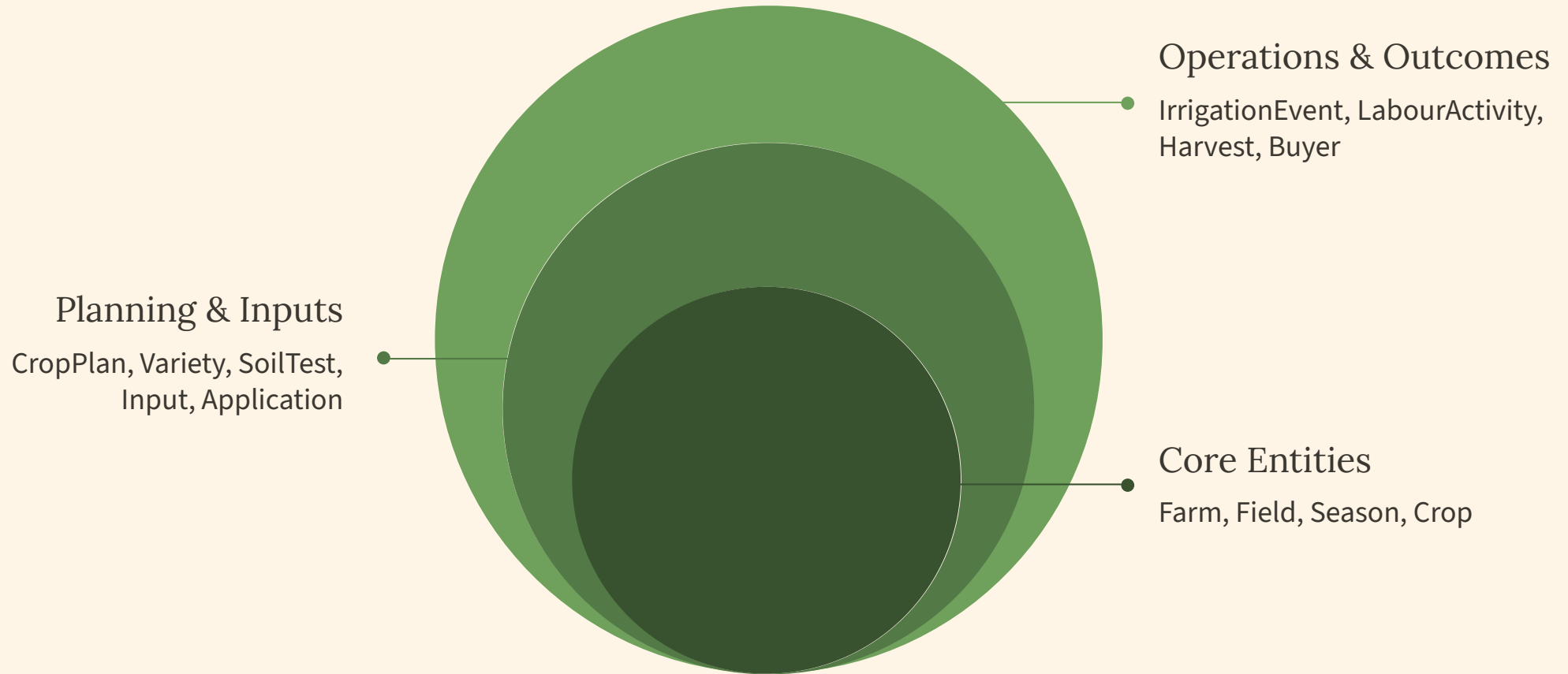
Yield, quality, storage



Buyer / Sale

Transactions and
pricing

Conceptual ER Diagram



A structured ER diagram showing cardinalities and primary foreign-key flows (Farm→Field→Season→CropPlan → activities → Harvest → Sale → Buyer).

Initial Relational Schema (Core tables)

Table	Primary / Key FKs (summary)
Farm	farm_id (PK); owner, location
Field	field_id (PK); farm_id (FK)
Season	season_id (PK); field_id (FK); start_date, end_date
CropPlan	plan_id (PK); season_id (FK); crop_id (FK); targets
Input / Application	input_id (PK); plan_id (FK); application records
IrrigationEvent	irrigation_id (PK); plan_id (FK); event_time
LabourActivity	labour_id (PK); plan_id (FK); hours, task
Harvest	harvest_id (PK); plan_id (FK); yield, quality
Buyer / Sale	sale_id (PK); harvest_id (FK); buyer_id (FK); amount

Focus on the most important 8–10 tables with clear PK–FK links; additional lookup tables (crop, variety, input types) planned.

Business Rules & Roadmap

1

Business Rules (preview)

Crop dates must fall within season; inputs & applications must be positive and within crop duration; harvest and sales non-negative and within stock.

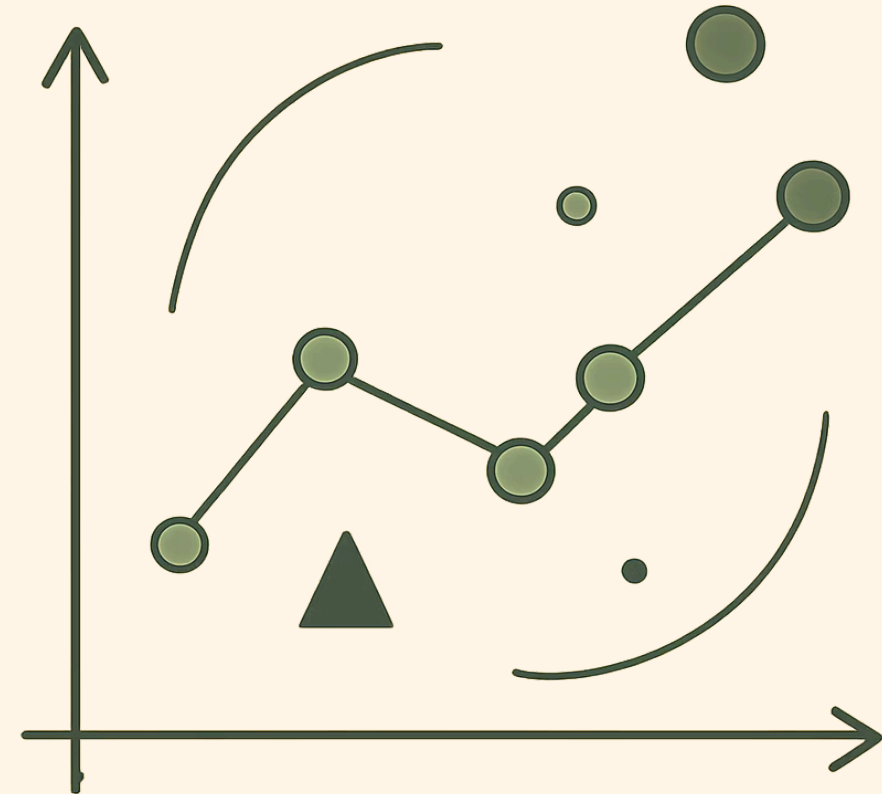
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Roadmap

Review 1 (Week 7): requirements, ER, initial schema — done.

Review 2 (Week 11): 3NF, DDL, sample queries.

Review 3 (Week 16): front-end app, CRUD, final demo.



📌 Next steps: normalize to 3NF, implement DDL and sample queries, prototype front-end for data entry and reports.

Thank you — Questions?

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